

Gaurav Singh

Full-Stack & AI Software Engineer | Distributed Systems & Machine Learning

+91 7019804464 | gaurav.workspace007@gmail.com | linkedin.com/in/grvsnh | github.com/grvsnh | grvsnh.dev | Bangalore, IN

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C++, SQL, HTML5/CSS3, Bash

Frontend & UI: React, Next.js, Electron, Three.js, GSAP, HTML5 Canvas, WebGL, Streamlit

Backend & APIs: Node.js, FastAPI, RESTful APIs, Microservices, OpenRouter API, WebRTC, Web Speech API

AI & Machine Learning: TensorFlow, TensorFlow.js, OpenCV, Deep Learning (CNNs), Grad-CAM, Pandas, Plotly.js

Databases & Developer Tools: PostgreSQL, Redis, IndexedDB, Docker, Git/GitHub, Linux (Bash), Postman, CI/CD

Operating Systems: Linux (Arch Linux, Ubuntu, Debian), macOS, Windows

EXPERIENCE

Web Developer Intern

Sep 2025 – Oct 2025

SkillCraft Technology

India, Remote

- Developed responsive web applications using modern HTML, CSS, and JavaScript, following industry best practices.
- Completed a series of hands-on development tasks and projects, implementing interactive UI components and responsive layouts.
- Strengthened front-end development, debugging, and problem-solving skills by building and refining real-world web solutions.

PROJECTS

Aiely | Electron, Node.js, OpenRouter API, Web Speech API

- Engineered a cross-platform AI desktop copilot for macOS and Windows with a native floating overlay that remains hidden during screen sharing and recordings.
- Built a real-time context engine combining screen analysis with dual-channel speech recognition (microphone + system audio) to generate instant coding assistance and meeting responses.
- Designed a multi-model failover architecture using OpenRouter with local-only configuration and zero telemetry, ensuring reliable AI inference, enhanced privacy, and uninterrupted operation.

BrainScan | TensorFlow.js, JavaScript, HTML/CSS, IndexedDB

- Engineered a privacy-first, browser-based AI application for brain tumor classification using TensorFlow.js with WebGL-accelerated client-side inference.
- Built advanced visualization tools including Grad-CAM heatmaps, an interactive before/after comparison slider, and a magnification lens for enhanced diagnostic interpretability.
- Reduced repeat model load times by implementing IndexedDB caching for model weights, enabling fully serverless deployment and zero patient data transmission.

F1 Strategy Lab | Next.js, TypeScript, FastAPI, Python, Plotly.js

- Architected a full-stack telemetry analysis platform that processes official Formula 1 race data to deliver interactive performance and strategy insights.
- Implemented high-performance data visualizations including driver telemetry comparison, circuit mapping, speed heatmaps, and lap delta analytics using Plotly.js.
- Designed backend APIs and analytics modules for tyre strategy evaluation, pit stop analysis, and recommendation generation using FastAPI and FastF1.

PhotoBooth | React, Three.js, GSAP, HTML5 Canvas

- Engineered a real-time interactive graphics application using Three.js and GSAP, implementing mathematically interpolated curtain simulations and physics-inspired animation pipelines.
- Designed computational geometry and image-processing algorithms including AABB collision detection, procedural film grain synthesis, texture generation, and canvas-based photo compositing.
- Built a responsive React application integrating WebRTC camera APIs with GPU-accelerated rendering and optimized animation workflows for seamless performance across desktop and mobile devices.

CERTIFICATIONS

Harvard University: CS50's Introduction to Programming with Python

Deloitte Australia (Forage): Technology Job Simulation

Postman: API Fundamentals Student Expert (Canvas Credentials)

HackerRank: Python (Basic) Certification

EDUCATION

Bachelor of Engineering in Computer Science

Visvesvaraya Technological University

2027

Bangalore, IN

All India Senior School Certificate Examination (AISSCE)

Army Public School PRTC

2022

Bangalore, IN

All India Secondary School Examination (AISSE)

Army Public School

2020

India